THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS FULL SYLLABUS PAPER SET 04

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The distance of the point $(9, 12)$ from the origin is: 1

(a) 16 units **(b)** 9 units **(c)** 15 units **(d)** 21 units

2. A point lies on the y-axis at distance 5 units below the origin. Its coordinates are: 1

(a) $(-5, 0)$ **(b)** $(5, 0)$ **(c)** $(0, 5)$ **(d)** $(0, -5)$

3. The midpoint of the segment joining $A(-3, 5)$ and $B(7, -1)$ is: 1

(a) $(4, 4)$ **(b)** $(2, -2)$ **(c)** $(2, 2)$ **(d)** $(5, -3)$

4. For the line $y = 4x - 7$, the slope and the y-intercept are respectively: 1

(a) -7 and 4 **(b)** -4 and -7
(c) 4 and 7 **(d)** 4 and -7

5. A student claims: "$5x^2 - 3$ is a linear polynomial." The correct response is: 1

(a) False, its degree is 2 so it is quadratic
(b) True
(c) True only for $x > 0$
(d) Cannot be determined

6. The distance between -4 and 3 on the number line is: 1

(a) 12 **(b)** 7 **(c)** −7 **(d)** 1

7. The number $3 + \sqrt{5}$ is: 1

(a) irrational **(b)** rational
(c) neither rational nor irrational **(d)** not a real number

8. The absolute value $|-\frac{3}{5}|$ is: 1

(a) 0 **(b)** $\frac{3}{5}$ **(c)** $-\frac{3}{5}$ **(d)** $\frac{5}{3}$

9. The value of 105^2 using a suitable identity is: 1

(a) 11045 **(b)** 11125 **(c)** 11015 **(d)** 11025

10. Which of the following is an identity? 1

(a) $x + 5 = 9$ **(b)** $2x = 10$ **(c)** $(x + y)^2 = x^2 + 2xy + y^2$ **(d)** $x^2 - 1 = 24$

11. In a circle of radius 17 cm, a chord is 8 cm from the centre. The length of the chord is: 1

(a) 28 cm **(b)** 30 cm **(c)** 15 cm **(d)** 32 cm

12. The perpendicular bisector of a chord of a circle always passes through: 1

(a) the centre of the circle **(b)** the midpoint of the arc only
(c) no fixed point **(d)** a point outside the circle

13. A chord of length 30 cm is at a distance of 8 cm from the centre of a circle. The radius of the circle is: 1

(a) 30 cm **(b)** 23 cm **(c)** 18 cm **(d)** 17 cm

14. Taking $\pi = \frac{22}{7}$, the circumference of a circle of radius 21 cm is: 1

(a) 264 cm **(b)** 1386 cm **(c)** 66 cm **(d)** 132 cm

15. By Heron's formula, the area of a triangle with sides 8 cm, 15 cm and 17 cm is: 1

(a) 60 cm^2 **(b)** 54 cm^2 **(c)** 120 cm^2 **(d)** 66 cm^2

16. Āryabhaṭa called his value 3.1416 for π 'āsanna', meaning: 1

(a) exact **(b)** largest possible
(c) wrong **(d)** approximate (approaching)

17. The number of outcomes in the sample space when a coin is tossed and a die is rolled is: 1

(a) 6 **(b)** 14 **(c)** 12 **(d)** 24

18. An event occurred 6 times in 40 trials. Its experimental probability (relative frequency) is: 1

(a) 0.075 **(b)** 0.25 **(c)** 6.67 **(d)** 0.15

19. The 25th term of the AP $2, 6, 10, \ldots$ is: 1

(a) 99 **(b)** 98 **(c)** 97 **(d)** 196

20. Which term of the sequence $t_n = 7n + 2$ is 261? 1

(a) $n = 39$ **(b)** $n = 38$ **(c)** $n = 37$ **(d)** $n = 36$

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. Show that the points $A(1,\ 1)$, $B(3,\ 3)$ and $C(5,\ 5)$ are collinear using distances. 2

22. Find the equation of the line through $(0,\ 3)$ with slope 2. Where does it cut the y-axis? 2

23. Find two rational numbers between $\frac{1}{2}$ and $\frac{3}{4}$. 2

24. Show that the identity $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ holds for $a = 2,\ b = 1$. 2

25. Side BC of cyclic quadrilateral ABCD is produced to E and $\angle DCE = 70^\circ$. Find $\angle A$. 2

26. An isosceles triangle has equal sides 5 cm and base 6 cm. Find its area. 2

27. One month of the year is chosen at random. What is the probability that it has 31 days? 2

28. A taxi charges a fixed ₹200 and ₹40 per km. Write the fare after 1, 2, 3 km and find the fare for 10 km. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Given $A(2,\ -2)$ and $B(8,\ 4)$, find the point P on AB that divides it in the ratio $1 : 2$ (nearer to A) and verify that $AP = \frac{1}{3}AB$. 3

30. Draw (describe the points for) the graphs of $y = \frac{1}{2}x$, $y = x$ and $y = 2x$ using suitable points. What does the value of a in $y = ax$ tell about the line? 3

31. Find the lengths of the first three hypotenuses in the square-root spiral starting with a right triangle of legs 1, 1. 3

32. Saira arranges a square of side x, 8 strips $x \times 1$ and 15 unit squares into a rectangle. Find its length and breadth. 3

33. Show that Heron's formula is a special case of Brahmagupta's formula. 3

34. Distinguish between subjective and objective (experimental/theoretical) probability using the example 'it will rain today'. 3

35. ABCD is cyclic, $\angle BAC = 30^\circ$ and $\angle BCD = 100^\circ$. Find $\angle CBD$. 3

OR

An arc AB subtends $\angle AOB = (6x - 10)^\circ$ at the centre and $\angle ACB = (2x + 5)^\circ$ at a point C on the circle. Find x. 3

36. Show that the sum of two consecutive triangular numbers is a perfect square; verify for t_{10} and t_9. 3

OR

The points (n, t_n) of the AP $2, 5, 8, \ldots$ are plotted. Show they lie on a straight line and find its slope. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) Simplify $\left(\frac{1}{4} - \frac{5}{12}\right) \div \frac{2}{3}$.
(ii) Find the sum and the product of $\frac{3}{5}$ and $\frac{7}{10}$. 5

OR

(i) Find the rational number x such that $x\left(\frac{1}{3} + \frac{1}{6}\right) = \frac{9}{2}$. 5
(ii) Verify $\frac{1}{2}\left(\frac{3}{4} + \frac{1}{3}\right) = \frac{1}{2} \times \frac{3}{4} + \frac{1}{2} \times \frac{1}{3}$.

38. Prove that a cyclic parallelogram is a rectangle and that its diagonals are diameters of the circle. 5

OR

Prove that if two chords of a circle are equidistant from the centre, they are equal; then find the length of each chord if the distance is 12 cm and the radius is 13 cm. 5

39. **Case study:** A circular park of radius 35 m has a circular fountain of radius 7 m at its centre. ($\pi = \frac{22}{7}$) 5
(i) Find the area of the park.
(ii) Find the area of the fountain.
(iii) Find the area of grass.
(iv) Find the length of railing needed to fence the fountain and the park boundary together.

OR

Case study: A pizza of radius 14 cm is cut into 8 equal sectors. ($\pi = \frac{22}{7}$) 5
(i) Find the angle of each sector.
(ii) Find the arc length of one slice.
(iii) Find the perimeter of one slice.
(iv) Find the area of one slice.

40. Classify each as AP, GP or neither, and give the nth term where it exists: 5

(a) $7, 11, 15, 19, \ldots$

(b) $2, 6, 18, 54, \ldots$

(c) $1, 4, 9, 16, \ldots$

(d) $100, 50, 25, 12.5, \ldots$

OR

(i) Plot the points (n, t_n) for the AP $1, 5, 9, 13, 17$ and say what you observe. 5

(ii) Plot the points for the GP $3, 6, 12, 24, 48$ and compare.

(iii) What does this tell you about growth in an AP and a GP?

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND